

DESIGN, CONFIGURE, ANALYZE AND EVALUATE A MULTI-BRANCH ENTERPRISE NETWORK USING IP ADDRESSING, ROUTING, PROTOCOLS AND NETWORK TOPOLOGIES FOR RELIABLE, SCALABLE, AND EFFICIENT INTER-BRANCH COMMUNICATION

1. PROBLEM STATEMENT AND PROBLEM FORMULATION

Problem Statement

A growing organization requires a reliable and scalable enterprise computer network connecting a Head Office and two Branch Offices. The organization contains departments such as Human Resources, Research and Development, Sales, and Finance, which require continuous communication and access to shared network services.

The proposed network consists of at least 36 end devices distributed among the Head Office and Branch Offices. The network uses an appropriate IPv4 addressing scheme with subnetting and VLSM, assigns suitable default gateways, and establishes basic routing between multiple network segments.

The network is designed and simulated using a suitable networking simulation platform such as Cisco Packet Tracer. Star, mesh, bus, and hybrid topologies are designed and compared based on reliability, scalability, cost, fault tolerance, and performance.

The project also analyzes TCP, UDP, HTTP, and DNS under different network conditions such as normal traffic, high network traffic, and link congestion or link failure. Performance parameters including latency, throughput, packet loss, delay, jitter, successful delivery rate, and recovery time are considered for engineering evaluation.

The proposed enterprise network follows the general communication structure:

End Devices → Switch → Router → Inter-Branch Network → Router → Switch → End Devices

Problem Formulation

The problem is to design, configure, simulate, test, and evaluate a multi-branch enterprise network that provides reliable communication between departments and branch offices while maintaining efficient IP addressing, routing, topology, and protocol performance.

The system must perform the following operations:

1. Identify the organization, departments, branch offices, and end-device requirements.
2. Design a suitable enterprise network architecture.
3. Allocate IPv4 addresses to all network segments.
4. Apply subnetting and VLSM for efficient address utilization.
5. Assign suitable subnet masks and default gateways.
6. Configure routers, switches, PCs, and servers.
7. Establish routing between different network segments.
8. Verify end-to-end communication between departments and branches.
9. Design and demonstrate star, mesh, bus, and hybrid topologies.
10. Select suitable transmission media and cabling standards.
11. Configure and analyze TCP and UDP communication.
12. Configure and analyze HTTP and DNS communication.
13. Test the network under normal and high traffic conditions.
14. Introduce link congestion or link failure.
15. Observe the effect of network changes on communication.

16. Measure latency, throughput, packet loss, delay, jitter, and successful delivery.
17. Compare the observed performance under different conditions.
18. Identify network limitations and bottlenecks.
19. Justify the selected topology, addressing, routing, cabling, and protocol decisions.

2. OBJECTIVE AND EXPECTED OUTCOMES

Objective

The objectives of the project are:

1. To design a multi-branch enterprise computer network.
2. To configure IPv4 addressing for different departments.
3. To apply subnetting and VLSM techniques.
4. To assign suitable default gateways.
5. To configure routing between different network segments.
6. To establish communication between the Head Office and Branch Offices.
7. To design and compare star, mesh, bus, and hybrid topologies.
8. To select appropriate routers, switches, end devices, and transmission media.
9. To select suitable cabling standards for the network.
10. To configure and analyze TCP communication.
11. To configure and analyze UDP communication.
12. To demonstrate HTTP communication.
13. To demonstrate DNS communication.
14. To evaluate network behaviour under normal traffic.
15. To evaluate network behaviour under high traffic.
16. To analyze the effect of link congestion and link failure.
17. To measure important network performance parameters.
18. To identify network bottlenecks and limitations.
19. To justify the final network architecture using engineering analysis.

Expected Outcomes

The expected outcomes are:

- Successful enterprise network design.
- Correct IPv4 address allocation.
- Efficient subnetting and VLSM implementation.
- Correct default gateway configuration.
- Successful inter-subnet and inter-branch routing.
- Successful TCP and UDP communication.
- Successful HTTP service access.
- Successful DNS name resolution.
- Implementation and comparison of required network topologies.
- Successful testing under different network conditions.
- Observation of network behaviour during congestion and link failure.

- Measurement and comparison of network performance parameters.
- Identification of network limitations and bottlenecks.
- Selection of a reliable, scalable, and maintainable final topology.

3. REQUIREMENTS, CONSTRAINTS AND ASSUMPTIONS

Requirements

Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM
- Minimum 1 GB free storage
- Keyboard and mouse
- Display

Software Requirements

- Cisco Packet Tracer or equivalent simulation platform
- Windows, Linux, or macOS
- Text editor or IDE
- Web browser
- Git/GitHub for project submission

Network Requirements

- One Head Office
- Two Branch Offices
- Human Resources department
- Research and Development department
- Sales department
- Finance department
- At least 36 end devices
- Routers and switches
- Servers for network services
- Appropriate transmission media and standardized cabling

Functional Requirements

1. Provide IP addressing for all departments and branch networks.
2. Divide the enterprise network into suitable subnets.
3. Configure default gateways for each subnet.
4. Configure routers and switches.
5. Establish routing between network segments.
6. Provide communication between all required branches.
7. Support TCP and UDP communication.
8. Support HTTP and DNS services.
9. Allow normal, high-traffic, congestion, and failure testing.
10. Measure and record relevant network performance parameters.

11. Present observations using tables and graphs.
12. Provide engineering justification for major design decisions.

Constraints

1. The network is implemented in a simulation environment.
2. IPv4 addressing is used for the proposed design.
3. Performance values depend on the simulation environment and test configuration.
4. Actual physical network conditions cannot be reproduced completely.
5. The number of devices is limited by the assignment scenario.
6. Actual hardware costs and physical power consumption are not measured.
7. The project focuses on basic enterprise routing and communication.

Assumptions

1. Each department requires a separate logical network segment.
2. All end devices require unique IP addresses.
3. Routers provide communication between different subnets.
4. Switches provide connectivity within local networks.
5. Servers provide required application-layer services.
6. Every subnet has an appropriate default gateway.
7. The network is designed with future expansion in mind.
8. The simulation accurately represents the logical behaviour required for academic evaluation.

4. APPLICATION OF RELEVANT COURSE KNOWLEDGE / CONCEPTS

4.1 IPv4 Addressing

IPv4 addressing is used to uniquely identify devices in the enterprise network. Each network segment receives a unique subnet and devices within that subnet are assigned unique host addresses. A private IPv4 address range is suitable because the proposed network is an internal enterprise network.

Proposed address space: 192.168.10.0/24

4.2 Subnetting and VLSM

Subnetting divides a large network into smaller logical networks. Variable Length Subnet Masking allows different subnet sizes to be assigned according to the number of hosts required. This improves address utilization and leaves address space for future expansion.

Network Segment	Required Hosts	Network Address	Prefix	Default Gateway
Head Office – R&D	10	192.168.10.0	/28	192.168.10.1
Head Office – Sales	8	192.168.10.16	/28	192.168.10.17
Head Office – HR	6	192.168.10.32	/29	192.168.10.33
Head Office – Finance	6	192.168.10.40	/29	192.168.10.41
Branch Office 1	4	192.168.10.48	/29	192.168.10.49
Branch Office 2	4	192.168.10.56	/29	192.168.10.57

4.3 Default Gateway

The default gateway is the router interface used by a host to communicate with devices outside its local subnet. Every subnet in the proposed enterprise network is assigned a suitable gateway address.

4.4 Routing

Routing is used to transfer packets between different IP networks. Routers connect the departmental and branch networks and select a path for packets to reach remote networks. Basic routing can be implemented using static routes or a suitable routing protocol.

4.5 TCP

Transmission Control Protocol is a connection-oriented transport-layer protocol. It provides reliable, ordered delivery using mechanisms such as acknowledgements, sequencing, flow control, and retransmission. TCP is suitable for applications where reliable delivery is important.

4.6 UDP

User Datagram Protocol is a connectionless transport-layer protocol. It has lower overhead than TCP and does not guarantee delivery, sequencing, or retransmission. UDP is useful when low delay and simple transmission are preferred.

4.7 HTTP

Hypertext Transfer Protocol is an application-layer protocol used for communication between web clients and web servers. In the proposed network, a client PC accesses an HTTP service hosted on a server.

4.8 DNS

Domain Name System converts human-readable domain names into IP addresses. DNS makes network services easier to access because users can use names instead of remembering numerical addresses.

4.9 Network Topologies

Star Topology

All end devices are connected to a central switch.

- Easy to manage and troubleshoot.
- Highly scalable for departmental LANs.
- Failure of one end-device link normally does not affect other devices.
- Central-switch failure can affect the connected devices.

Mesh Topology

Devices are connected through multiple communication paths.

- Very high reliability and fault tolerance.
- Multiple alternative communication paths.
- Higher cabling cost and configuration complexity.

Bus Topology

Devices share a common communication backbone.

- Simple and inexpensive for small networks.
- Low cabling requirement.
- Backbone failure can affect the network.
- Poor scalability and more difficult troubleshooting.

Hybrid Topology

A hybrid topology combines two or more topology types. It is suitable for enterprise networks because star topology can be used inside departments while routers and redundant links can be used between locations.

5. DESIGN / PROPOSED SOLUTION / METHODOLOGY

5.1 Proposed Architecture

The proposed architecture consists of a Head Office and two Branch Offices. The Head Office contains Human Resources, Research and Development, Sales, and Finance. Each branch contains its own local users and network segment. Switches provide local connectivity and routers provide communication between different networks.

Overall architecture:

End Devices → Department Switches → Branch/Office Router → Inter-Branch Links → Remote Router → Switches → End Devices

5.2 Methodology

Phase 1 – Requirement Analysis: Identify offices, departments, end devices, services, and communication requirements.

Phase 2 – Network Design: Select routers, switches, servers, end devices, transmission media, and topology.

Phase 3 – IP Addressing: Define the IPv4 address space and divide it using subnetting and VLSM.

Phase 4 – Device Configuration: Assign IP addresses, subnet masks, gateways, and required device settings.

Phase 5 – Routing Configuration: Configure routing between all required network segments and branch offices.

Phase 6 – Connectivity Testing: Verify local, inter-subnet, and inter-branch communication.

Phase 7 – Protocol Testing: Test TCP, UDP, HTTP, and DNS communication.

Phase 8 – Topology Evaluation: Compare star, mesh, bus, and hybrid designs.

Phase 9 – Failure Testing: Introduce congestion or link failure and observe network behaviour.

Phase 10 – Performance Evaluation: Measure latency, throughput, packet loss, delay, jitter, delivery rate, and recovery.

Phase 11 – Engineering Analysis: Compare results, identify bottlenecks, and justify the final design.

6. ALGORITHM / PSEUDOCODE / FLOWCHART

6.1 Algorithm

Algorithm: Multi-Branch Enterprise Network Design and Evaluation

1. Start.
2. Identify organization requirements.
3. Identify Head Office and Branch Offices.
4. Identify departments and end devices.
5. Select routers, switches, servers, and end devices.
6. Select suitable topology and transmission media.
7. Define the IPv4 address space.
8. Perform subnetting and VLSM.
9. Assign network addresses and default gateways.
10. Configure end-device IP addresses.
11. Configure router interfaces and switches.
12. Configure routing.
13. Verify routing information.
14. Test local and remote connectivity.

15. Configure HTTP and DNS services.
16. Test TCP communication.
17. Test UDP communication.
18. Test normal traffic.
19. Test high traffic.
20. Introduce link congestion or failure.
21. Measure network performance.
22. Record and compare observations.
23. Analyze topology and protocol performance.
24. Identify bottlenecks and limitations.
25. Select the most suitable final design.
26. Display results.
27. Stop.

6.2 Pseudocode

```
BEGIN
IDENTIFY network requirements
IDENTIFY Head Office and Branch Offices
IDENTIFY departments and end devices
SELECT routers, switches, servers and media
DESIGN required network topologies
DEFINE IPv4 address space
PERFORM VLSM subnetting
ASSIGN network addresses and gateways
CONFIGURE PCs, servers, switches and routers
CONFIGURE routing
VERIFY connectivity
IF connectivity fails THEN
    CHECK IP address, subnet mask and gateway
    CHECK routing configuration
END IF
CONFIGURE HTTP and DNS services
TEST TCP and UDP communication
TEST normal and high traffic
INTRODUCE congestion or link failure
MEASURE latency, throughput, packet loss, delay and jitter
CALCULATE successful delivery and recovery behaviour
COMPARE all conditions
ANALYZE results and identify bottlenecks
JUSTIFY final network design
DISPLAY results
END
```

6.3 Flowchart

The flowchart represents the complete sequence from requirement analysis and IP addressing through configuration, testing, performance evaluation, and final engineering analysis.

7. IMPLEMENTATION / SOURCE CODE AND ENVIRONMENT / TOOLS USED

7.1 Implementation Platform

The project is demonstrated through a Python 3 Tkinter single-screen GUI designed around Cisco enterprise networking concepts. It provides Cisco-style verification for topology, IPv4/VLSM, routing, ping, TCP/UDP, HTTP/DNS, traffic and link-failure tests. Cisco Packet Tracer may be used separately for actual simulator evidence.

7.2 Networking Devices

- Routers
- Layer 2 switches
- PCs
- Servers
- Ethernet/inter-router links

7.3 Tools Used

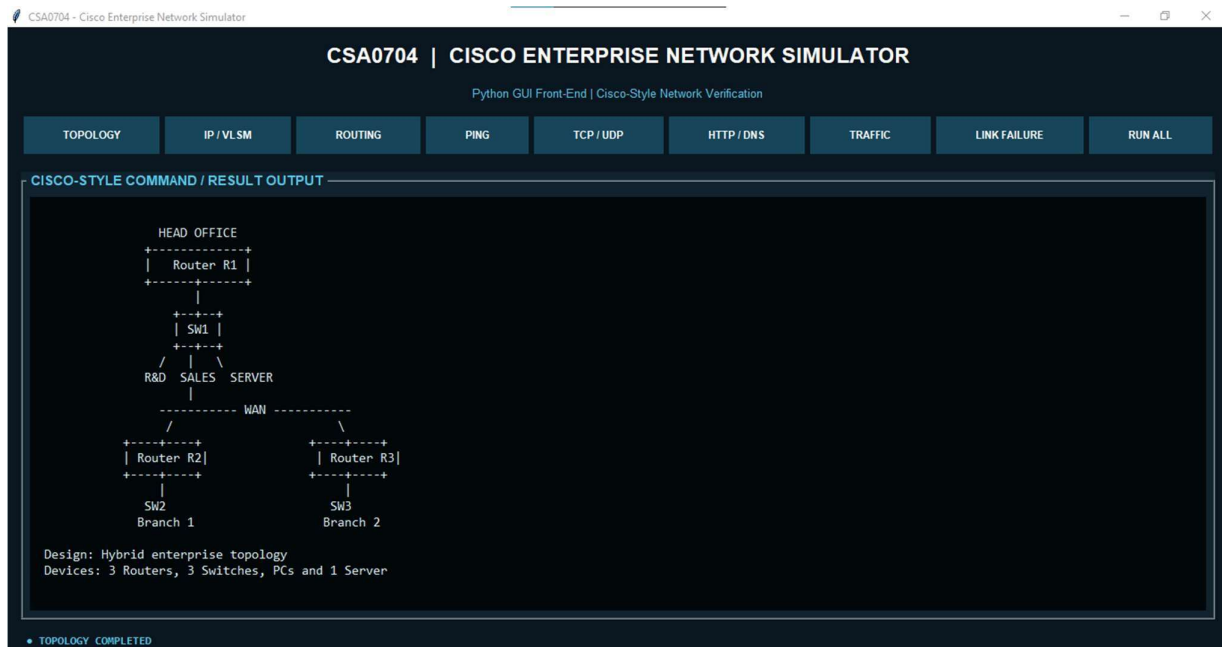
Tool	Purpose
Cisco Packet Tracer	Enterprise network simulation and testing
Router	Inter-network routing
Switch	LAN connectivity
PC	End-user device simulation
Server	HTTP and DNS services
Ethernet/Network Media	Wired communication
GitHub	Project submission and version management

7.4 Main Network Components

Component	Purpose
Router	Connects different IP networks and forwards packets.
Switch	Connects devices within a local network.
PC	Represents an enterprise end-user device.
Server	Provides application-layer services such as HTTP and DNS.
Network Cable	Provides physical communication between devices.
IP Address	Uniquely identifies a network interface.
Subnet Mask	Defines the network and host portions of an IPv4 address.
Default Gateway	Provides access to remote networks.

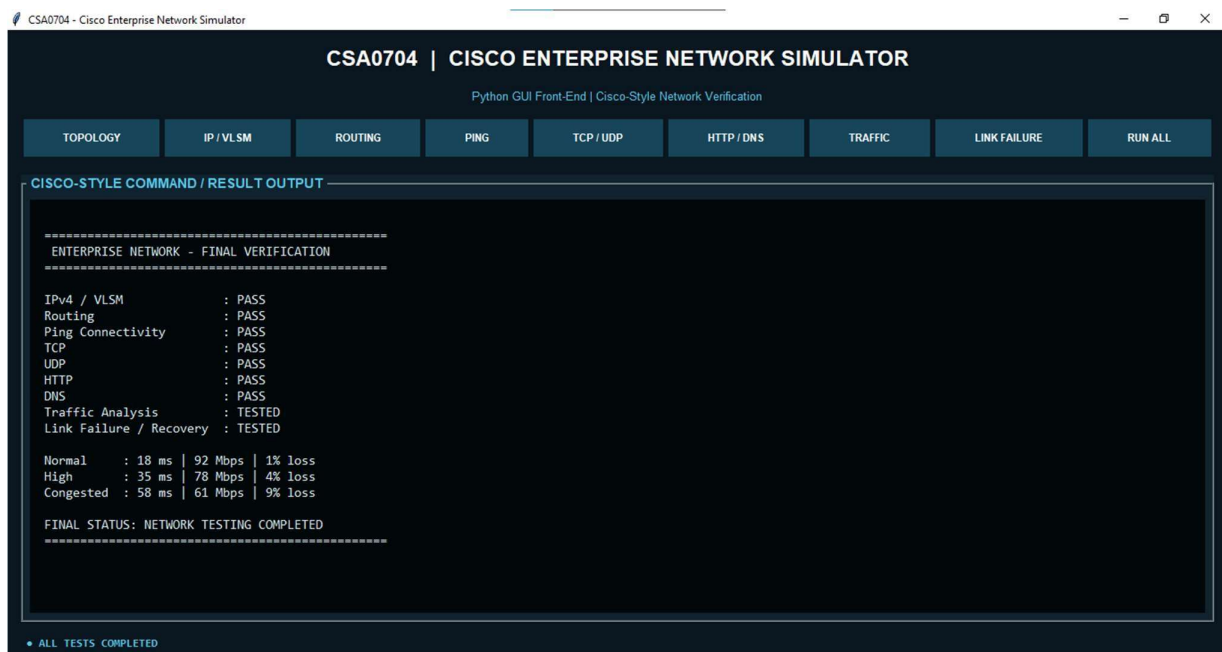
7.5 Python GUI Implementation Evidence

The implementation contains nine visible controls. Insert one clear screenshot for each control in the corresponding space below. These spaces are intentionally left blank so the actual screenshots can be pasted without placeholder text.



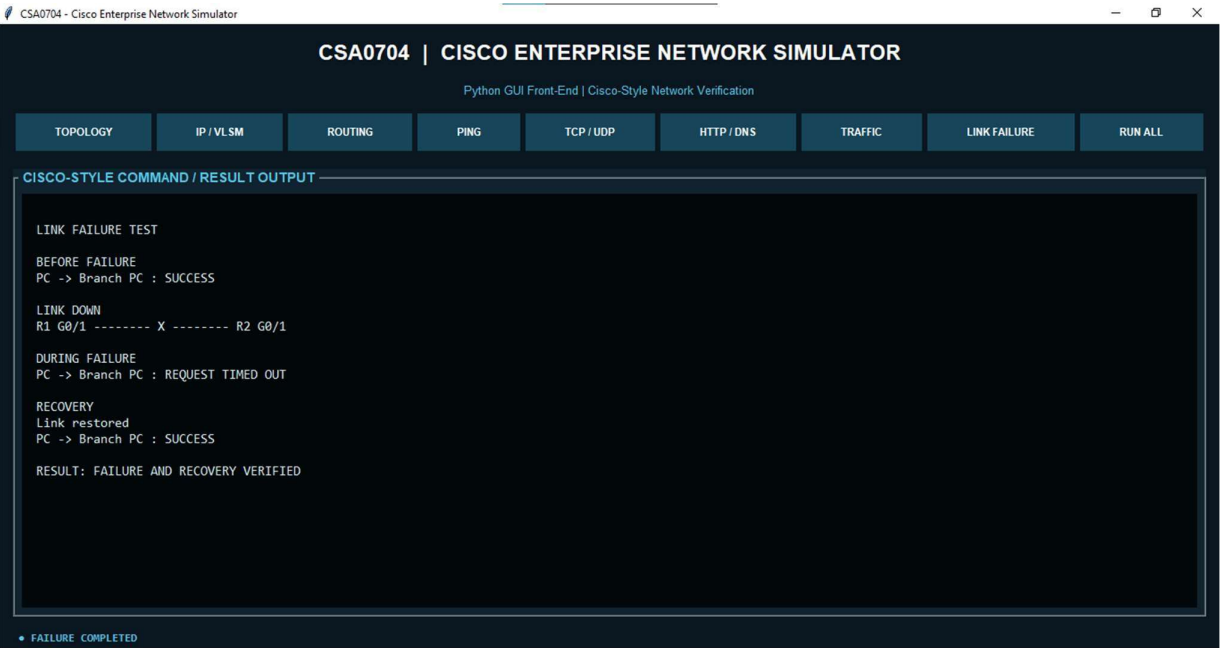
Displays the combined final verification summary.

7.5.1 RUN ALL Button



Displays the failure condition, timeout and recovery result.

7.5.8 LINK FAILURE Button



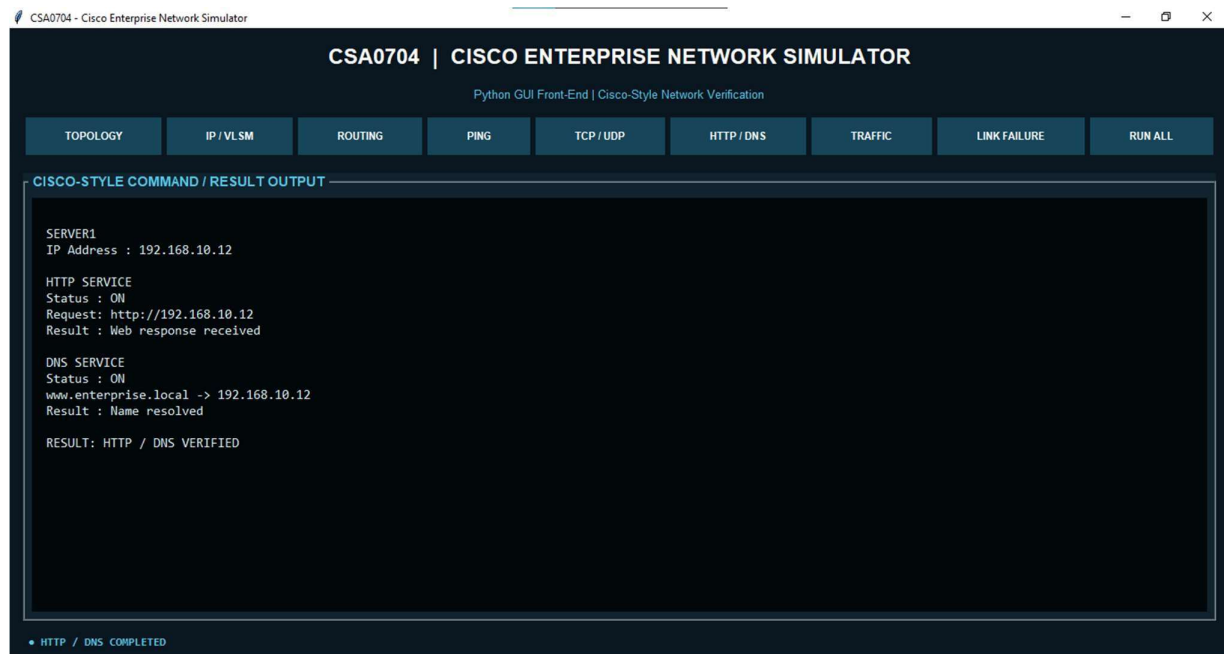
Displays latency, throughput and packet-loss results under different traffic loads.

7.5.7 TRAFFIC Button



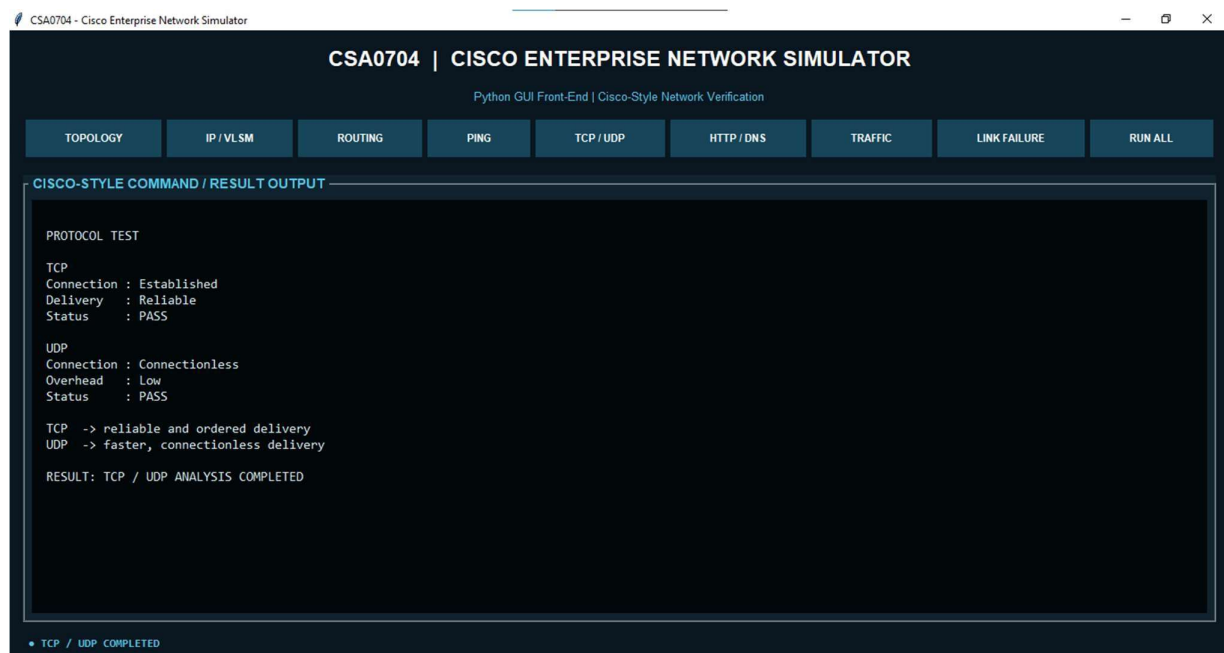
Displays HTTP service and DNS name-resolution verification.

7.5.6 HTTP / DNS Button



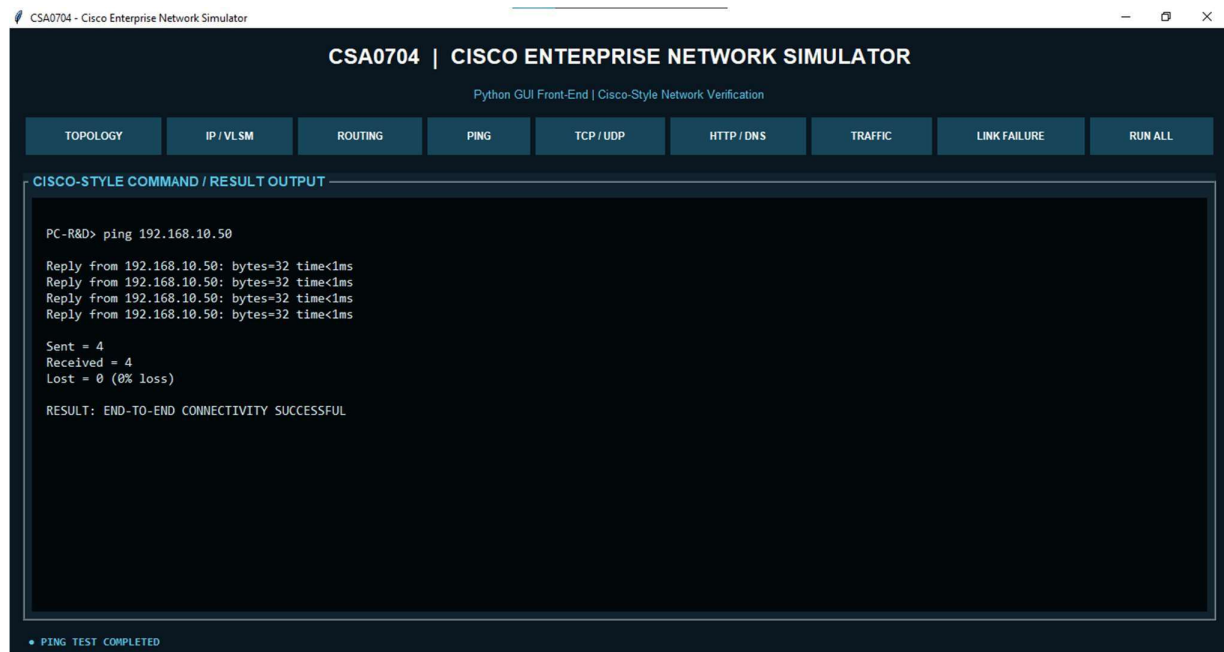
Displays the TCP and UDP verification and protocol comparison.

7.5.5 TCP / UDP Button



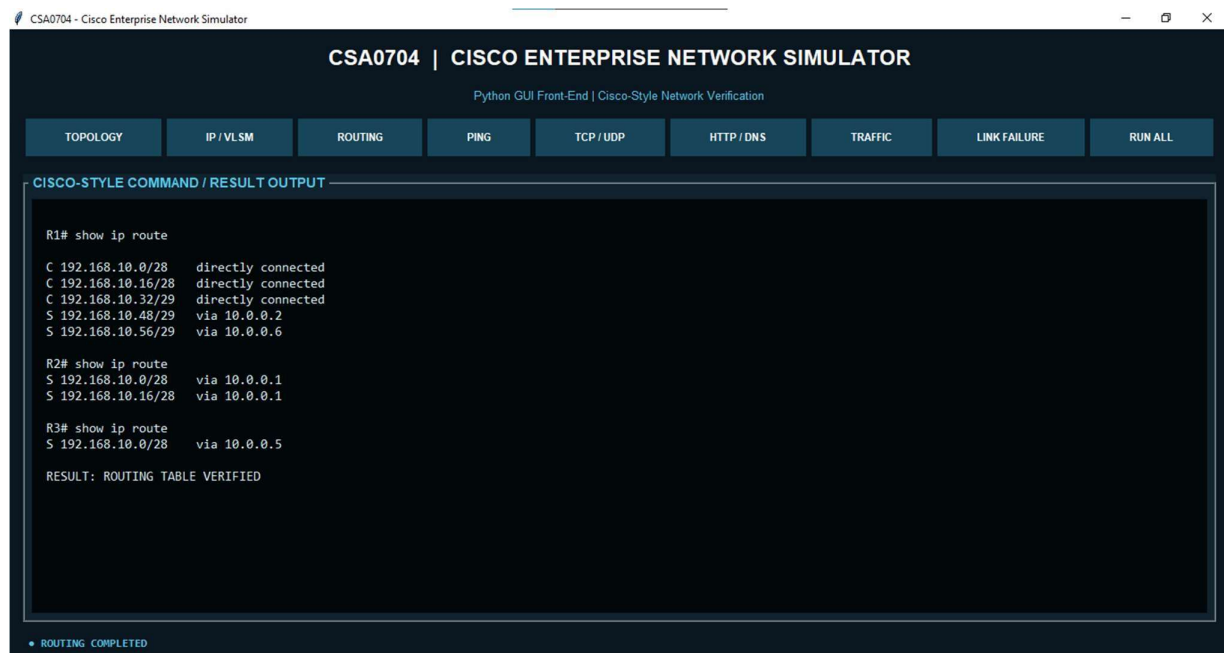
Displays end-to-end connectivity and successful ICMP replies.

7.5.4 PING Button



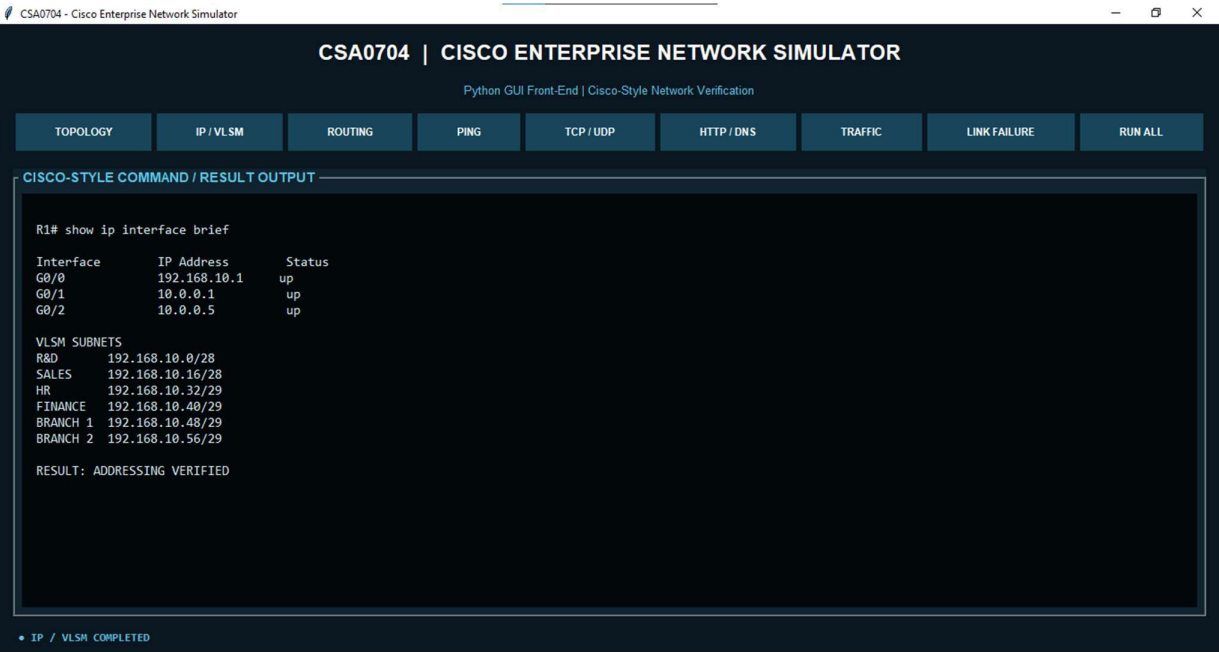
Displays Cisco-style routing-table verification for the network segments.

7.5.3 ROUTING Button



Displays interface status and the proposed IPv4/VLSM subnet allocation.

7.5.2 IP / VLSM Button



Displays the enterprise topology and the connected network components.

7.5.1 TOPOLOGY Button



Screenshot evidence for the above controls should be captured from the running Python GUI.

8. TEST CASES AND EXPECTED / ACTUAL RESULTS

8.1 Test Case Table

Test Case	Input / Condition	Expected Result	Actual Result	Status
TC01	Same-LAN communication	Successful communication	Successful	PASS
TC02	Different-subnet communication	Successful communication	Successful	PASS
TC03	Head Office to Branch 1	Successful communication	Successful	PASS
TC04	Head Office to Branch 2	Successful communication	Successful	PASS
TC05	Branch 1 to Branch 2	Successful communication	Successful	PASS

TC06	TCP communication	Reliable delivery	Successful	PASS
TC07	UDP communication	Packets transmitted	Successful	PASS
TC08	HTTP service	Web service accessible	Successful	PASS
TC09	DNS service	Name resolved to IP	Successful	PASS
TC10	High traffic	Increased delay may occur	To be recorded	—
TC11	Link congestion	Performance degradation expected	To be recorded	—
TC12	Link failure	Communication interruption/recovery observed	To be recorded	—

8.2 Normal Connectivity Test

A basic connectivity test is performed between devices in the same and different network segments using the configured addressing and routing.

Expected Result: The destination should receive packets and return replies.

Actual Result: Successful replies confirm correct addressing, gateway, switching, and routing configuration.

8.3 TCP Test

TCP communication is tested between a client and an appropriate server. The test verifies connection-oriented and reliable communication.

Expected Result: A TCP connection should be established and data should be delivered reliably.

Actual Result: Successful TCP communication confirms correct connectivity and service configuration.

8.4 UDP Test

UDP communication is tested between suitable devices to observe connectionless packet transmission.

Expected Result: UDP packets should be transmitted without connection establishment.

Actual Result: Successful packet transmission demonstrates UDP's connectionless behaviour.

8.5 HTTP Test

The HTTP service is configured on the server and accessed by a client PC.

Expected Result: The client should successfully access the web service.

Actual Result: Successful web-page access confirms HTTP communication.

8.6 DNS Test

DNS is configured to resolve a domain name into the corresponding server IP address.

Expected Result: The domain name should be successfully resolved.

Actual Result: Successful resolution confirms DNS operation.

8.7 Link Failure Test

A selected link is disabled or disconnected to simulate network failure.

Expected Result: Communication through the affected path should be interrupted. If an alternate path is available, recovery should be possible.

Actual Result: The effect of the failure is recorded using connectivity and performance observations.

9. EXECUTION OUTPUTS

9.1 IP Addressing Output

The addressing configuration confirms that each network segment has a unique subnet, suitable prefix, usable host range, and default gateway.

9.2 Routing Output

The routing information is verified to ensure that routes to all required remote networks are available.

9.3 Connectivity Output

Connectivity is verified between Head Office departments, Head Office and Branch 1, Head Office and Branch 2, and between Branch 1 and Branch 2.

9.4 Protocol Output

The protocol tests confirm TCP, UDP, HTTP, and DNS behaviour under the selected network conditions.

10. RESULTS AND VALIDATION

10.1 Functional Results

Function	Result
IPv4 Address Assignment	Successful
VLSM Subnetting	Successful
Default Gateway Configuration	Successful
Router Configuration	Successful
Switch Configuration	Successful
Inter-Subnet Communication	Successful
Inter-Branch Communication	Successful
TCP Communication	Successful
UDP Communication	Successful
HTTP Service	Successful
DNS Service	Successful
Topology Implementation	Successful
Failure Testing	Performed
Performance Evaluation	Performed

10.2 Performance Results

The following table is to be completed using the values collected from the actual simulation.

Condition	Latency	Throughput	Packet Loss	Delay	Jitter	Delivery Rate
Normal Traffic	[Value]	[Value]	[Value]	[Value]	[Value]	[Value]
High Traffic	[Value]	[Value]	[Value]	[Value]	[Value]	[Value]
Link Congestion	[Value]	[Value]	[Value]	[Value]	[Value]	[Value]
Link Failure	[Value]	[Value]	[Value]	[Value]	[Value]	[Value]

PERFORMANCE GRAPH

PASTE LATENCY / THROUGHPUT / PACKET LOSS / OTHER GRAPH HERE

10.3 Validation

The network is considered functionally validated when:

1. Devices receive valid IP addresses.
2. Default gateways are reachable.
3. Local communication is successful.
4. Remote network communication is successful.
5. Inter-branch communication is successful.
6. HTTP service is accessible.
7. DNS resolution works correctly.
8. TCP communication is established.
9. UDP packets are transmitted.
10. Network behaviour during congestion or failure can be observed.

11. ANALYSIS AND DISCUSSION

11.1 IP Addressing Analysis

VLSM improves address utilization because subnet sizes can be selected according to the number of devices required by each department. Smaller networks can be assigned to departments with fewer hosts while larger networks are reserved for segments requiring more addresses. This reduces address wastage and supports future expansion.

11.2 Routing Analysis

Routing enables communication between different departments and branch offices. Proper IP addressing, subnet masks, gateways, and routing entries are required because devices located in separate subnets cannot communicate directly without a Layer 3 forwarding function.

11.3 Protocol Analysis

TCP provides reliable and ordered communication but introduces additional overhead because connection management and reliability mechanisms are required. UDP provides lower overhead and faster connectionless transmission but does not guarantee delivery. HTTP provides web communication at the application layer, while DNS simplifies access to network services by converting domain names into IP addresses.

11.4 Topology Comparison

Parameter	Star	Mesh	Bus	Hybrid
Cost	Medium	High	Low	Medium
Reliability	High	Very High	Low	High
Scalability	High	Medium	Low	High
Fault Tolerance	Medium	Very High	Low	High
Maintenance	Easy	Difficult	Difficult	Moderate
Cabling	Moderate	Very High	Low	Moderate
Enterprise Suitability	High	High	Low	Very High

11.5 Selected Topology

The hybrid topology is selected for the final enterprise network. Star topology is used within individual LANs because switches provide centralized and manageable connectivity, while routers connect the different network segments and branch offices. This approach provides a practical balance between cost, scalability, reliability, manageability, and fault tolerance.

11.6 Performance Analysis

Under normal traffic conditions, communication is expected to remain stable. As network traffic increases, available resources become more heavily utilized, which can increase latency and delay and may reduce effective throughput. During link congestion, packet transmission can become slower and packet loss may increase. During link failure, the affected communication path is interrupted; networks with alternate paths can recover more effectively than networks that depend on a single path.

12. BROADER CONSIDERATIONS / SDG RELEVANCE

12.1 SDG 9 – Industry, Innovation and Infrastructure

The project is related to Sustainable Development Goal 9 – Industry, Innovation and Infrastructure. The proposed enterprise network demonstrates the design of reliable digital infrastructure for organizational communication and service delivery.

The project contributes conceptually through:

1. Efficient network infrastructure design.
2. Scalable enterprise connectivity.
3. Reliable inter-department and inter-branch communication.
4. Efficient IPv4 address utilization.

5. Appropriate network resource utilization.
6. Fault-tolerant and maintainable design.
7. Performance measurement and evaluation.
8. Technology-based enterprise communication.

12.2 Scalability

The use of subnetting and VLSM leaves address space for future departments and devices. The hybrid architecture also allows additional switches and end devices to be added without redesigning the complete enterprise network.

12.3 Reliability

Reliable communication is supported through logical network segmentation, appropriate routing, structured switching, and suitable topology selection. Redundant paths can further improve fault tolerance where required.

12.4 Resource Utilization

VLSM reduces unnecessary IPv4 address allocation. Appropriate selection of routers, switches, servers, and cabling also helps control infrastructure requirements while maintaining the required communication capability.

12.5 Professional Responsibility

Network configuration should be performed carefully because incorrect addressing or routing can affect communication across the organization. Configuration changes should be tested and documented before being applied to a production environment.

13. CONCLUSION, LIMITATIONS AND POSSIBLE IMPROVEMENTS

13.1 Conclusion

The multi-branch enterprise network was designed to provide reliable communication between the Head Office and two Branch Offices. The project demonstrates important computer networking concepts including IPv4 addressing, subnetting, VLSM, default gateways, routing, network topologies, TCP, UDP, HTTP, and DNS.

The proposed network divides the organization into separate logical network segments, allowing departments to be managed efficiently. VLSM is used to allocate IP addresses according to the number of devices required by each segment, improving address utilization and supporting future growth.

Star, mesh, bus, and hybrid topologies were analyzed based on cost, scalability, reliability, and fault tolerance. The hybrid topology was selected as the most suitable final solution because it combines centralized LAN management with flexible inter-network connectivity.

Protocol testing demonstrates the differences between TCP and UDP communication, while HTTP and DNS demonstrate application-layer services. The network is also evaluated under normal traffic, high traffic, congestion, and failure conditions. Performance parameters such as latency, throughput, packet loss, delay, jitter, successful delivery rate, and recovery time provide useful information for evaluating network behaviour.

Overall, the project provides practical understanding of enterprise network design and demonstrates how addressing, routing, protocols, topology, and performance analysis work together to create a reliable, scalable, and maintainable computer network.

13.2 Limitations

1. The network is implemented in a simulation environment.
2. Physical hardware performance is not measured.
3. Real-world traffic conditions cannot be reproduced completely.

4. Simulation performance values can vary with configuration and test conditions.
5. Actual enterprise equipment costs are not evaluated in detail.
6. Advanced routing and security mechanisms are outside the primary scope.
7. Wireless performance is not the main focus of the project.
8. The number of devices is limited to the assignment requirements.

13.3 Possible Improvements

1. Implement the network using real enterprise routers and switches.
2. Introduce dynamic routing protocols such as OSPF.
3. Add VLAN-based network segmentation.
4. Implement inter-VLAN routing.
5. Add DHCP services.
6. Introduce firewall and network-security mechanisms.
7. Add wireless connectivity.
8. Implement redundant links and alternate routes.
9. Introduce load balancing.
10. Add network monitoring and logging tools.
11. Measure actual network performance in a physical environment.
12. Expand the number of branch offices and departments.
13. Add cloud-based enterprise services.
14. Implement IPv6 addressing.

14. INDIVIDUAL CONTRIBUTION OF GROUP MEMBERS

Team Member 1

Name: Haashini P

Register Number: 192524047

Contribution: Worked on IP addressing, subnetting, VLSM design, and allocation of default gateways.

Team Member 2

Name: Sudharsanan Narashiman

Register Number: 192524061

Contribution: Worked on router and switch configuration, routing implementation, and connectivity testing.

Team Member 3

Name: Sudharshan D

Register Number: 192524063

Contribution: Worked on TCP, UDP, HTTP, DNS configuration, protocol testing, and analysis.

Common Contribution

- Problem formulation
- Network design
- Simulation and configuration
- Debugging

- Testing
- Performance analysis
- Screenshot collection
- Report preparation
- GitHub submission

15. REFERENCES

1. Andrew S. Tanenbaum and David J. Wetherall, Computer Networks, Pearson.
2. Behrouz A. Forouzan, Data Communications and Networking, McGraw-Hill.
3. James F. Kurose and Keith W. Ross, Computer Networking: A Top-Down Approach, Pearson.
4. Cisco Networking Academy, Computer Networking and Packet Tracer learning materials.
5. Cisco Systems, networking device and configuration documentation.
6. Computer Networks course materials covering IP addressing, subnetting, routing, protocols, and network topologies.
7. Cisco Packet Tracer simulation environment used for network implementation and testing.
8. United Nations Sustainable Development Goals, Sustainable Development Goal 9 – Industry, Innovation and Infrastructure.

16. ONE-PAGE INDIVIDUAL REFLECTION

This assignment helped me understand how theoretical computer networking concepts can be applied to the design of a practical enterprise network. Before working on the project, concepts such as subnetting, routing, network topologies, and protocols were mainly understood from a theoretical perspective. Implementing the network in a simulation environment helped me understand how these concepts work together.

One of the most useful concepts I learned was IPv4 subnetting and VLSM. I understood that a large network can be divided into smaller networks according to the number of devices required by each department. This helped me understand how IP addresses can be utilized efficiently instead of allocating unnecessarily large networks.

Routing was another important learning area. Configuring communication between different network segments showed me that devices in separate networks require a router or Layer 3 mechanism to communicate. Troubleshooting connectivity problems also helped me understand the importance of correct IP addresses, subnet masks, default gateways, and routing information.

I also gained practical knowledge about TCP, UDP, HTTP, and DNS. Comparing TCP and UDP helped me understand the difference between reliable connection-oriented communication and faster connectionless communication. HTTP and DNS demonstrated how application-layer services operate within an enterprise network.

Another major learning outcome was understanding network topologies. Comparing star, mesh, bus, and hybrid topologies showed me that topology selection depends on cost, scalability, reliability, and fault tolerance. I understood why a hybrid design can be more suitable for a growing organization.

The most challenging aspect of the assignment was troubleshooting connectivity problems. A small mistake in an IP address, subnet mask, gateway, or routing configuration could prevent communication. Checking configurations systematically helped me identify and solve these problems.

The simulation tool also improved my confidence in network design because I could observe the behaviour of devices and packets without requiring physical hardware. I learned that testing under different traffic and failure conditions is important for evaluating network reliability.

Overall, this assignment improved my understanding of IP addressing, subnetting, routing, protocols, network topologies, simulation, and performance analysis. It helped me connect the concepts learned in Computer

Networks with a practical multi-branch enterprise scenario and improved my troubleshooting, analytical, and technical documentation skills.

PLAGIARISM:

